Theory

ISPhO

A1-1

Korean (Korea (Republic of))

강자성 도메인벽 동역학

Part A. 단일 자기 모멘트 (2.0점)

A.1 (0.3pt)

A.2 (0.9pt)

$m_z(t) =$

A.3 (0.8pt)

$\Omega =$

Part B. 정적 도메인벽 (3.5점)

B.1 (0.5pt)

(식) $B_{\text{ex}} =$

(값) $B_{\text{ex}} =$

(식) $B_{\text{an}} =$

(값) $B_{\text{an}} =$

(식) $\omega =$

(값) $\omega =$

B.2 (1.0pt)

$\alpha =$

$T =$

$d =$

B.3 (1.5pt)

β	d

B.4 (0.5pt)

$c =$

Part C. 외부 자기장 속 도메인벽 운동 (4.5점)

C.1 (0.2pt)

방향 (오른쪽 / 왼쪽):

C.2 (2.0pt)

b	v	φ

C.3 (0.4pt)
그래프:



$dv/db =$

C.4 (0.6pt)
그래프:



$v_c =$ $b_{\mathrm{cr}} =$

C.5 (0.3pt)

$v_c =$ $B_{\mathrm{cr}} =$

C.6 (1.0pt)

t	x

그래프:



Theory

ISPhO

A2-1

Korean (Korea (Republic of))

폴리인의 진동 구조

Part A. 무한한 탄소 원자 사슬 (6.5점)

A.1 (0.8pt)

$E =$

A.2 (0.8pt)

$m\ddot{u}_n =$

$m\ddot{v}_n =$

A.3 (0.9pt)

$a =$ $b =$

$c =$ $d =$

A.4 (1.0pt)

$\omega_1/\omega_0 =$

$\omega_2/\omega_0 =$

A.5 (1.0pt)

$\omega_1/\omega_0 \approx$

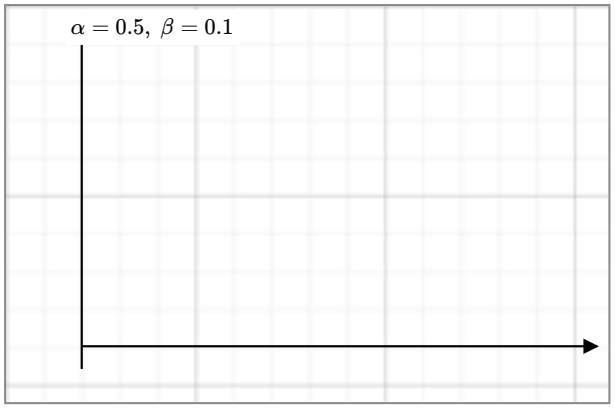
$\omega_2/\omega_0 \approx$

A.6 (1.0pt)

$\omega_1/\omega_0 \approx$

$\omega_2/\omega_0 \approx$

A.7 (1.0pt)
그래프:





Part B. 폴리인의 분광학 (3.5점)

B.1 (0.8pt)

식:

B.2 (0.7pt)

$q_l =$

B.3 (2.0pt)

단일 결합:

삼중 결합:

단전자 트랜지스터

Part A. 터널 효과 (3.0점)

A.1 (0.1pt)

$W =$ $\Delta W =$

A.2 (0.3pt)

$q_g =$ $q_j =$

A.3 (0.3pt)

$\Delta E =$

A.4 (0.5pt)

$A =$ $Q =$

A.5 (0.5pt)

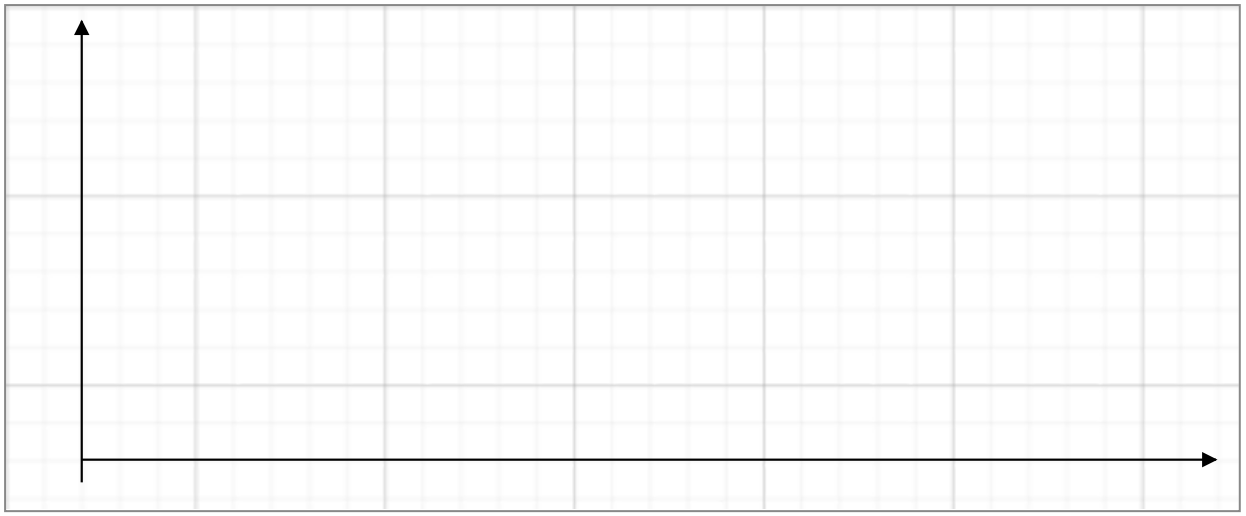
$I_t =$

n에 대한 조건:

A.6 (0.3pt)

$I =$

A.7 (1.0pt)
그래프:



Part B. 단전자 트랜지스터 (4.5점)

B.1 (0.2pt)

$\varphi =$

B.2 (1.0pt)

$A =$

B.3 (1.0pt)

$Q =$

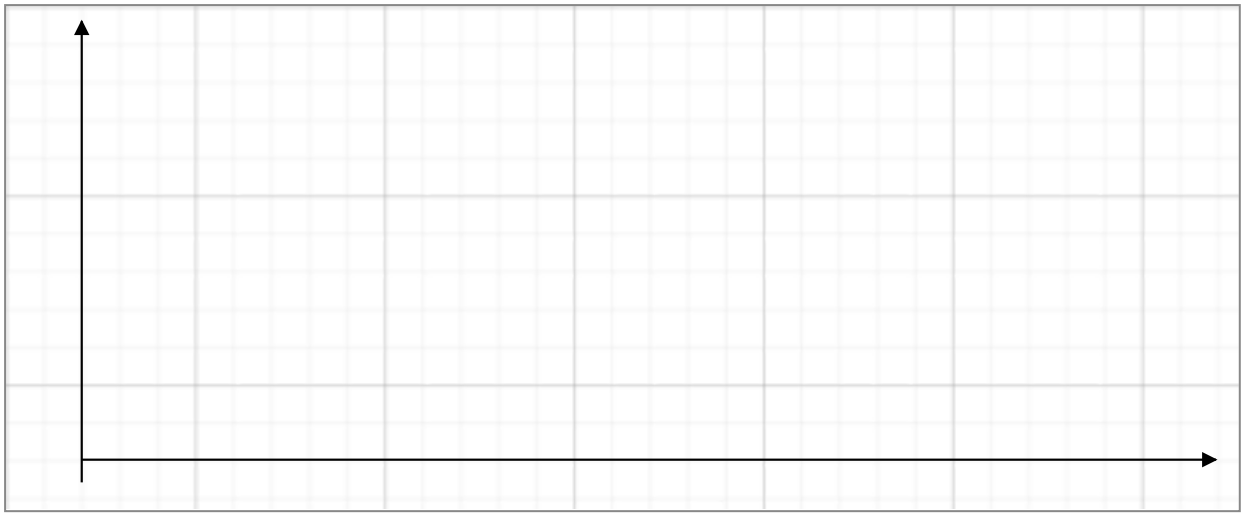
B.4 (0.3pt)

조건:

B.5 (1.5pt)

조건:

그래프:



B.6 (0.5pt)

$I(V_g, V) =$

Part C. 유한 온도에서의 터널링 (2.5점)

C.1 (0.5pt)

$C_1 = C_2 =$ $C_g =$

C.2 (0.3pt)

$dp(n)/dt =$

C.3 (0.5pt)

$p(n+1) =$

C.4 (0.3pt)

C.5 (0.5pt)

$p(0) =$ $p(1) =$

C.6 (0.4pt)

$I =$